



Computer Networks 04 | Introduction to Subnetting | CS & IT | GATE Crash Course

Generated on October 23, 2024

Summary

AI Notes

AI Slides

Text Notes

Screenshots

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QUESTION

Organization X need = 2²² IP addresses

PW

ANSWER



$X \rightarrow 2^{22} \checkmark$

$A \rightarrow 2^{22} \checkmark$

$B \rightarrow 2^{22} \checkmark$

$C \rightarrow 2^{22} \checkmark$

$4 \times 2^{22} = 2^3 \times 2^{22} = 2^{25}$

IP addresses wasted = $2^{24} - 2^{22}$

$$= 2^{22} \times 2^2 - 2^{22}$$

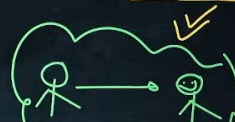
$$= 4 \times 2^{22} - 2^{22}$$

$$= 3 \times 2^{22}$$

$$= 3 \times 2^{22} \times 2^{20}$$

$$= 12 \times 2^{20}$$

$$= 12M$$

$$= \underline{\underline{12,582,912}}$$


▶ 24:29

II: Organization Y need = 2^{14} IP addresses | IP addresses wasted = $2^{16} - 2^{14}$

$= 2^2 * 2^{14} - 2^{14}$
 $= 4 * 2^{14} - 2^{14}$
 $= 3 * 2^{14}$
 $= 3 * 2^4 * 2^{10}$
 $= 48 * 2^{10}$
 $= 48K$
 $= 49,152$

Diagram: A circle divided into four quadrants. The top-right quadrant is labeled "class-B" and " $2^{16}IP$ ". The bottom-left quadrant is labeled "class-A" and " $2^{16}IP$ ". The bottom-right quadrant is labeled "class-C" and " $2^{16}IP$ ". The top-left quadrant is labeled "class-D" and " $2^{16}IP$ ".

▶ 25:05

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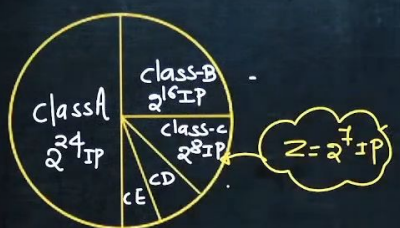
$Y = 2^{14}$
 $A = 2^{14}$
 $B = 2^{14}$
 $C = 2^{14}$
 $4 * 2^{14} = 2^2 * 2^{14} = 2^{16}$

Diagram: A circle divided into four quadrants. The top-left quadrant is labeled "Y" and " 2^{14} ". The top-right quadrant is labeled "A" and " 2^{14} ". The bottom-left quadrant is labeled "B" and " 2^{14} ". The bottom-right quadrant is labeled "C" and " 2^{14} ".

$\frac{2^{16}}{4} = \frac{2^{16}}{2^2} = 2^{14}$

▶ 28:14

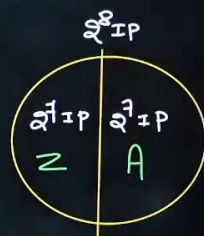
III: Organization Z need = 2^7 IP addresses | IP addresses wasted = $2^8 - 2^7$
 $= 128$



$$Z = 2^7$$

$$A = 2^7$$

$$\frac{2^8}{2} = 2^7$$



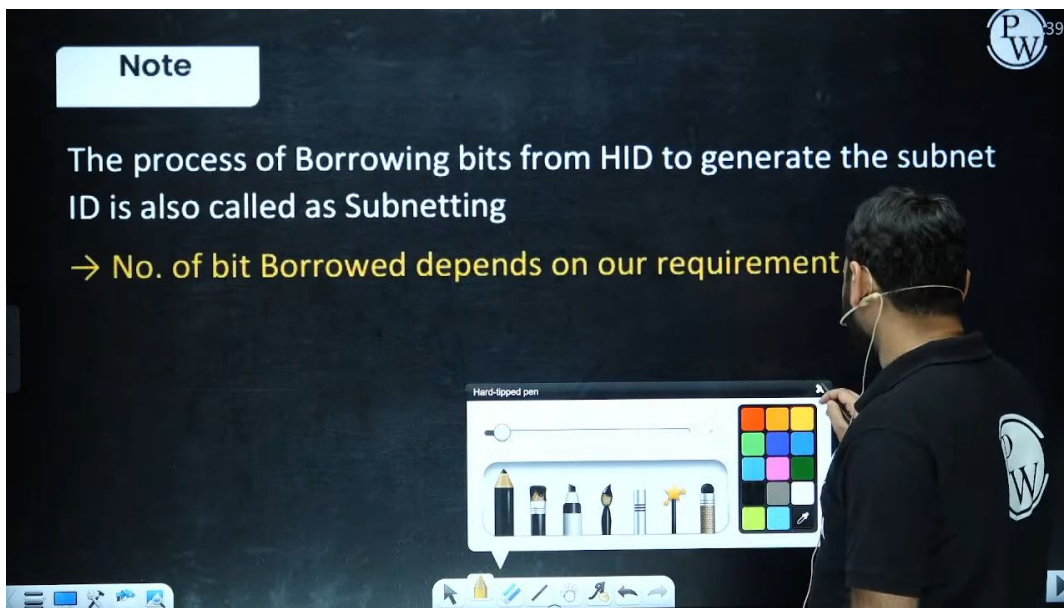
▶ 34:04

Subnetting

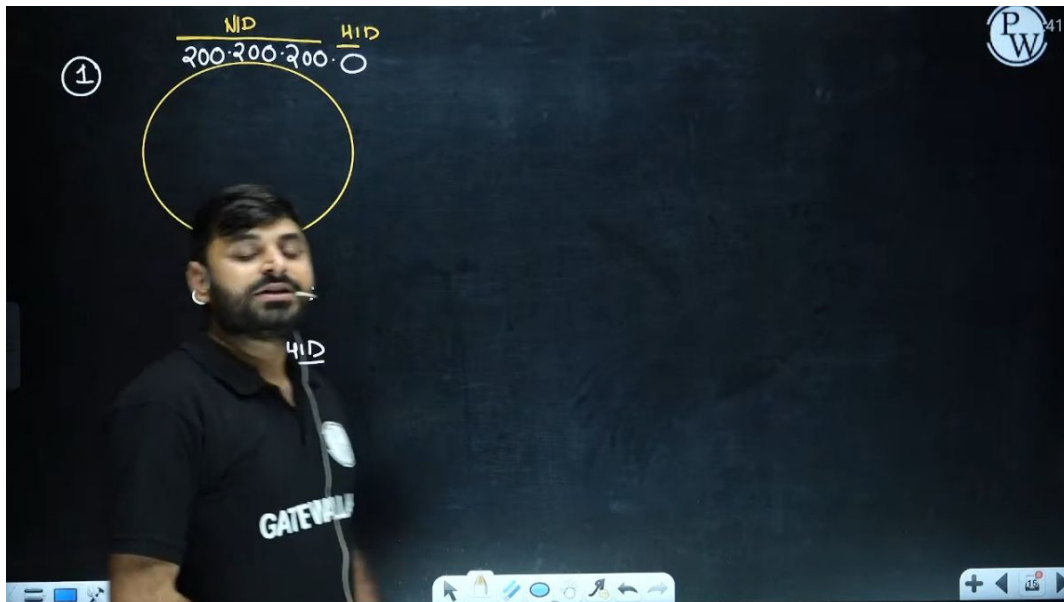
The process of dividing a big network to many smaller subnet is called as subnetting.



▶ 34:31



▶ 34:58



▶ 36:55

①

$\frac{\text{NID}}{200.200.200} \cdot \frac{\text{HID}}{0}$

$00 \rightarrow 1^{\text{st}} \text{ subnet}$
 $01 \rightarrow 2^{\text{nd}} \text{ subnet}$
 $10 \rightarrow 3^{\text{rd}} \text{ subnet}$
 $11 \rightarrow 4^{\text{th}} \text{ subnet}$

class-c

$\frac{\text{NID}}{24} \cdot \frac{\text{HID}}{8}$

4 subnet

$\frac{2}{\text{SID}} \cdot \frac{6}{\text{HID}} \rightarrow 2^6 - 2 \text{ Host/subnet}$

No. of subnets = $2^2 = 4$

▶ 44:20

1st subnet

$\frac{\text{NID}}{200.200.200} \cdot \frac{\text{SID}}{00} \cdot \frac{\text{HID}}{\text{-----}}$

$\frac{2}{\text{SID}} \cdot \frac{6}{\text{HID}}$

▶ 47:42

1st subnet

$\frac{200.200.200.00}{\text{NID}} \quad \frac{00}{\text{SID}} \quad \frac{\text{-----}}{\text{HID}}$

$\frac{200.200.200.00}{\text{NID}} \quad \frac{00}{\text{SID}} \quad 000000 \rightarrow 200.200.200.0 \text{] SID}$

$200.200.200.00 \quad 000000 \quad 1 \rightarrow 200.200.200.1 \text{] First Host}$

$200.200.200.00 \quad 000010 \rightarrow 200.200.200.2$

$\vdots$

$200.200.200.00 \quad 111110 \rightarrow 200.200.200.62 \text{] Last Host}$

$200.200.200.00 \quad 111111 \rightarrow 200.200.200.63 \text{] DBA}$

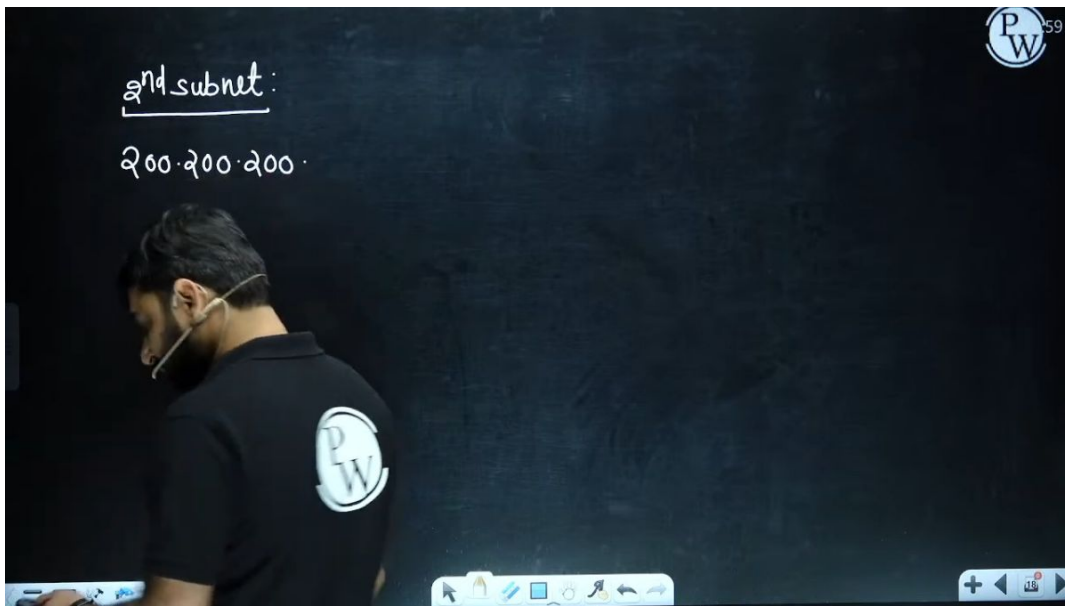
62 Hosts

▶ 54:35

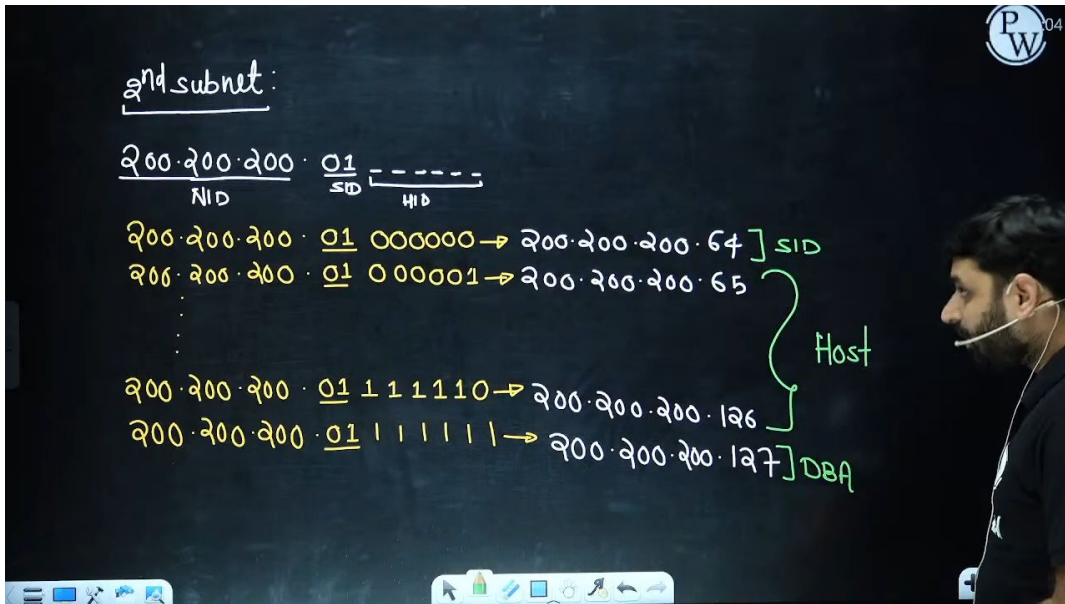
2nd subnet:

200.200.200.2

▶ 54:54



▶ 54:55



▶ 59:38

PW⁰⁷

3rd subnet

$\frac{200.200.200}{NID}$	$\frac{10}{SID}$	$\frac{-----}{HID}$	
---------------------------	------------------	---------------------	--

$200.200.200 \cdot \frac{10}{SID} 000000 \rightarrow 200.200.200.128] SID$

...

$200.200 \cdot \frac{10}{SID} 111111 \rightarrow 200.200.200.191] DBA$

$\begin{array}{r} 255 \\ 64 \\ \hline 191 \end{array}$

GATE WALLAH

▶ 1:03:02

PW¹⁰

4th subnet

$\frac{200.200.200}{NID}$	$\frac{11}{SID}$	$\frac{-----}{HID}$	
---------------------------	------------------	---------------------	--

$200.200.200 \cdot \frac{11}{SID} 000000 \rightarrow 200.200.200.132] SID$

...

$200.200 \cdot \frac{11}{SID} 111111 \rightarrow 200.200.200.255] DBA$

GATE WALLAH

▶ 1:05:14

PW¹⁹

shortcut

128 64
00 6 bit
HID

128 64
01 6 bit
HID

128 64
10 6 bit
HID

128 64
11 6 bit
HID

4th subnet: 200.200.200.192 - 200.200.200.255

1st subnet: SID: 200.200.200.0, DBA: 200.200.200.63

2nd subnet: SID: 200.200.200.64, DBA: 200.200.200.127

3rd subnet: SID: 200.200.200.128, DBA: 200.200.200.191

▶ 1:14:34

3. 200.200.200.0

ISRO

class-c

NID	HID
24	8

4 subnet

PW²⁷

$\frac{24}{NID}$	$\frac{6}{HID}$	$\frac{2}{SID}$
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✓ $\hookrightarrow$ No. of subnet = $2^2 = 4$

No. of Host/subnet = $2^6 - 2 = 62$

$\frac{2}{SID}$

00 $\rightarrow$ 1st subnet
 01 $\rightarrow$ 2nd "
 10 $\rightarrow$ 3rd "
 11 $\rightarrow$ 4th "

▶ 1:22:49

1st subnet

200.200.200.00000000 → 200.200.200.0
 NID HID SID

200.200.200.00000000 → 200.200.200.0 SID

200.200.200.00000001 → 200.200.200.1

200.200.200.00000010 → 200.200.200.2

⋮

200.200.200.11111000 → 200.200.200.248

200.200.200.11111000 → 200.200.200.252 DBA

▶ 1:29:09

6bit 21
 HID SID

6bit 21
 HID SID

6bit 21
 HID SID

6bit 21
 HID SID

1st subnet
 SID : 200.200.200.0
 DBA : 200.200.200.252

2nd subnet
 SID : 200.200.200.1
 DBA : 200.200.200.253

3rd subnet
 SID : 200.200.200.2
 DBA : 200.200.200.254

4th subnet
 SID : 200.200.200.3
 DBA : 200.200.200.255

▶ 1:34:10

Subnet Mask



It is a 32 bit number used to indicate no. of bits borrowed from host-id and there positions based on the following rules

Rule1: No of 1's in the subnet mask indicate NID+SID

Rule2: No of 0's in the subnet mask indicate HID part

▶ 1:36:01

Q.1

If NID = 200.200.200.0 and the subnet Mask = 255.255.255.192 then identify



No. of bit borrowed from Host-id → 2

No. of ⁴ subnet possible and their subnet id's

No. of Host/subnet (62)

So/7: 255.255.255.192
||||| . ||||| . ||||| . ||000000
NID SID HID

No. of 1's = 26
NID + SID = 26
24 + SID = 26
SID = 2 bit
No. of subnet = $2^2 = 4$

No. of 0's = 6
HID = 6 bit
No. of Host/subnet
= $2^6 - 2 = 62$

▶ 1:50:52

Q.2 If NID = 200.200.200.0 and the subnet Mask = 255.255.255.224 then identify

i. No. of bit borrowed from Host-id $\rightarrow 3$

ii. No. of subnet possible and their subnet id's

iii. No. of Host/subnet 30

Summary:

NID: 11111111.11111111.11111111.00000000
 No. of 1's = 27
 NID + SID = 27
 24 + SID = 27
 SID = 3 bit

SID: 000000
 No. of 0's = 5
 HID = 5 bit
 No. of Host/subnet = $2^5 - 2 = 30$

SID: Address

SID	Address
000	0
001	32
010	64
011	96
100	128
101	160
110	192
111	224

▶ 1:57:49

Q.3 If NID = 200.200.200.0 and the subnet Mask = 255.255.255.44 then identify

i. No. of bit borrowed from Host-id (3)

ii. No. of subnet possible and their subnet id's

iii. No. of Host/subnet (30)

Summary:

NID: 11111111.11111111.11111111.00101100
 No. of 1's = 27
 No. of 0's = 5

SID: Address

SID	Address
000	0
001	4
010	8
011	12
100	16
101	20
110	24
111	28

▶ 2:03:06

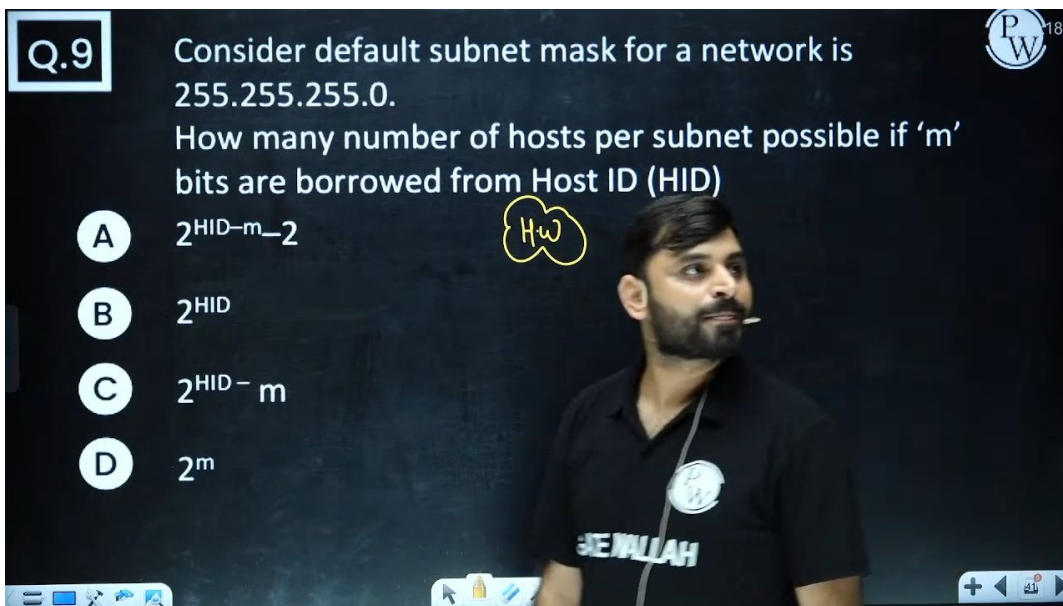
Q.9 Consider default subnet mask for a network is 255.255.255.0.
How many number of hosts per subnet possible if 'm' bits are borrowed from Host ID (HID)

A $2^{\text{HID}-m}-2$ H-w

B 2^{HID}

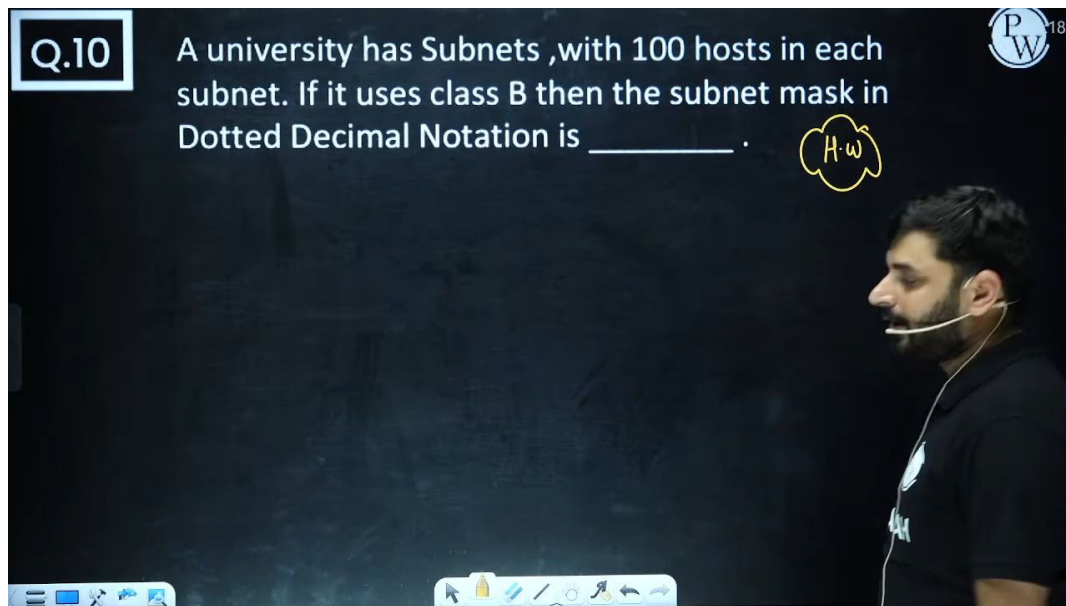
C $2^{\text{HID}-m}$

D 2^m



▶ 2:13:23

Q.10 A university has Subnets ,with 100 hosts in each subnet. If it uses class B then the subnet mask in Dotted Decimal Notation is _____ . H-w



▶ 2:13:36

▶ 2:14:43

